

Linda, 48 · 8 hours of **severe epigastric pain radiating to the back** · nausea, vomiting · lipase 1,200 (~12× normal) · ultrasound: gallstones in the gallbladder, dilated common bile duct, edematous pancreas · **gallstone pancreatitis**

BIO 004 · WEEK 5 · CASE DEEP DIVE

The **Accessory** Digestive Organs

A gallstone that escaped, lodged at the duodenum, and turned a digestive organ against itself.

PATIENT CASE

Linda is 48. After a large dinner last night she woke at 4 AM with **severe epigastric pain radiating straight through to her back**, with nausea and vomiting. She comes to the ED at 9 AM, doubled over.

Her abdomen is tender in the epigastrium and right upper quadrant. **Lipase is 1,200** (more than ten times normal); ALT and AST are mildly elevated. Right upper quadrant ultrasound shows **multiple stones in the gallbladder**, a **dilated common bile duct**, and an **edematous pancreas**. The diagnosis is **acute gallstone pancreatitis** — a stone passed from the gallbladder, lodged at the ampulla, and blocked pancreatic outflow.

The accessory digestive organs (liver, gallbladder, pancreas, salivary glands) sit OUTSIDE the alimentary canal but deliver secretions into it through ducts. When one duct system blocks the other, you get a case like Linda's.

GUIDING QUESTION

Where do the accessory digestive organs sit, how do their ducts converge on the duodenum, and what happens when a gallstone blocks the shared pancreatic-biliary outlet?

PREWORK

Companion to your Week 5 Accessory Digestive Organs prework page. Terms, video, and spaced recall for liver, gallbladder, pancreas, salivary glands, and the biliary tree live there.

Three pairs, three jobs

● Parotid (largest)

In front of the ear. Drains via Stensen's duct opposite the upper 2nd molar. Mostly serous. The gland inflamed in mumps.

● Submandibular

Below mandible. Drains via Wharton's duct under the tongue. Mixed serous and mucous.

● Sublingual

Under the tongue. Multiple small ducts. Mostly mucous.

Saliva: water, mucus, salivary amylase (starts carbohydrate digestion), lysozyme (antibacterial), bicarbonate (buffers acid). ~1-1.5 L per day.

The body's largest internal organ, with too many jobs to count

The liver sits in the right upper quadrant under the diaphragm, weighs about 1.5 kg, has four anatomic lobes (**right, left, caudate, quadrate**) and eight functional segments (each with its own portal triad branch).

The functional unit is the **hepatic lobule**: hexagonal in cross-section, with a central vein in the middle and a portal triad (hepatic artery branch, portal vein branch, bile duct) at each corner. Blood flows from portal triads through *sinusoids* lined by hepatocytes to the central vein.

Synthesis

Albumin, clotting factors (II, VII, IX, X), most plasma proteins. Bile.

Detoxification

Drugs, hormones, ammonia → urea.

Metabolism

Glucose storage as glycogen; gluconeogenesis; lipid handling.

Storage

Glycogen, vitamins A, D, B12, iron.

Bile production

~1 L per day. Emulsifies fats. Carries bilirubin (from broken-down hemoglobin) for excretion.

The plumbing that delivers bile to the duodenum

Bile leaves hepatocytes into bile canaliculi → bile ductules → right and left hepatic ducts → **common hepatic duct**. The **cystic duct** from the gallbladder joins the common hepatic duct to form the **common bile duct**. The common bile duct descends behind the duodenum, joins the **pancreatic duct**, and the two empty into the duodenum at the **ampulla of Vater (hepatopancreatic ampulla)**, guarded by the **sphincter of Oddi**.

Linda's problem in one sentence. A stone from the gallbladder slipped down the cystic duct, lodged at or just above the ampulla, and is blocking both bile flow AND pancreatic outflow. Bile backs up into the liver (elevated ALT/AST, dilated common bile duct on ultrasound). Pancreatic juice can't drain; trapped enzymes activate inside the pancreas and start digesting it.

Concentrator and reservoir for bile

Pear-shaped sac on the inferior surface of the liver. Holds 30-50 mL of bile. Mucosa concentrates bile up to 10× by reabsorbing water and ions. When the duodenum sees fat, enteroendocrine cells release *CCK*, which makes the gallbladder contract and the sphincter of Oddi relax → bile squirts into the duodenum to emulsify the fat.

Gallstones form when bile becomes supersaturated with cholesterol or bilirubin. Risk factors: **Four F's** — Female, Forty, Fertile, Fat. Most are silent until they obstruct the cystic duct (biliary colic), the common bile duct (cholangitis, jaundice), or — like Linda's — the ampulla (gallstone pancreatitis).

Two organs in one body

The pancreas lies retroperitoneal across the back of the abdomen, head nestled in the C of the duodenum, body and tail extending to the spleen. It is both **exocrine** (98% by mass) and **endocrine**.

Exocrine acini

Cluster of cells producing pancreatic juice: bicarbonate (neutralizes gastric acid in duodenum), *amylase* (starch), *lipase* (fat), *proteases as zymogens* (trypsinogen, chymotrypsinogen — activated only in the duodenum).

Islets of Langerhans

1-2 million scattered through the pancreas. *Alpha cells*: glucagon. *Beta cells*: insulin. *Delta cells*: somatostatin. Hormones go straight to the bloodstream.

The pancreatic duct runs the length of the gland and joins the common bile duct at the ampulla of Vater (in most people). An accessory pancreatic duct (duct of Santorini) drains separately in some people.

Why a blocked duct destroys the gland

Pancreatic enzymes are stored as inactive zymogens specifically to prevent autodigestion. They are activated only when they reach the duodenum. When pancreatic outflow is blocked, enzymes accumulate, activate prematurely inside the pancreas, and begin digesting the gland.

Three top causes of acute pancreatitis (mnemonic: **I GET SMASHED**): **Gallstones** (most common in women), **Ethanol** (most common in men), and a long tail including ERCP, trauma, scorpion sting, mumps, autoimmune, steroids, hypercalcemia, hypertriglyceridemia, drugs.

DRAW ZONE 1 -- Diagram the biliary tree from hepatocyte to duodenum. Label hepatic ducts, common hepatic duct, cystic duct, gallbladder, common bile duct, pancreatic duct, ampulla of Vater, sphincter of Oddi. Mark Linda's stone.

Star the place a stone produces gallstone pancreatitis (at or above the ampulla, blocking both ducts).



DRAW ZONE 2 -- Sketch a hepatic lobule in cross-section. Label central vein, portal triad (hepatic artery, portal vein, bile duct), sinusoids, hepatocytes. Show blood flow direction.

Six corners with portal triads, one central vein. Blood flows IN from triads, OUT through central vein; bile flows the opposite direction toward the triads.

One stone, two organs, every finding mapped

Linda's lab values and imaging are the accessory digestive system telling you exactly where the obstruction is and what damage it's causing.

"Severe epigastric pain radiating to back"

The pancreas is retroperitoneal. Inflammation irritates the retroperitoneum and refers pain to the back. Classic for pancreatitis.

"Lipase 1,200 (>3× normal)"

Pancreatic acinar cells are being damaged; lipase leaks into the bloodstream. >3× normal with consistent symptoms is diagnostic of acute pancreatitis.

"Elevated ALT/AST"

Bile backs up into the liver from the blocked common bile duct, irritating hepatocytes. ALT >3× normal in suspected pancreatitis strongly suggests gallstone etiology.

"Dilated common bile duct on ultrasound"

Confirms obstruction at or beyond the ampulla. Plus visible gallstones in the gallbladder = strong evidence for gallstone pancreatitis.

"Why the Four F's (Female, Forty, Fertile, Fat)"

Estrogen increases biliary cholesterol secretion. Obesity and rapid weight loss both promote stones. Fertile = elevated estrogen states (pregnancy, OCPs).

"Treatment plan"

Aggressive IV fluids, pain control, NPO, monitor for complications (necrotizing pancreatitis, ARDS, multi-organ failure). If stone doesn't pass: ERCP to extract it. Eventual cholecystectomy to prevent recurrence.

Liver, gallbladder, duct, pancreas. Linda's case is the accessory digestive plumbing failing at its convergence point.

